Binary Pattern Analysis Report

FFT, Periodicity, Entropy & Run-Length Diagnostics

Input file	bits.csv
Generated at	2026-05-18 14:05:52
Total samples	596
Detected values	Binary sequence: 0 / 1

Executive Summary

This report analyzes a binary sequence containing **596** samples. The sequence contains **388** ones and **208** zeros. The binary entropy is **0.9332** bits/symbol. The sequence shows a strong imbalance between zeros and ones. Entropy is moderate, suggesting some structure or symbol bias may exist. The sequence contains 175 groups of ones and 175 groups of zeros. The longest run of ones has length 186, while the longest run of zeros has length 34. FFT suggests a dominant period of approximately 2.01 samples. Autocorrelation highlights lag 2 as the strongest non-zero repeated structure.

1. Core Metrics

This section summarizes the most important high-level characteristics of the binary sequence.

Total samples **596** Number of binary observations analyzed.	**Ones** **388 (65.10%)** Share of symbols equal to 1.
Zeros **208 (34.90%)** Share of symbols equal to 0.	**Binary entropy** **0.9332** Maximum is 1.0 for a perfectly balanced binary source.
Bias vs 50/50 **+0.1510** Positive means more ones; negative means more zeros.	**Total runs** **350** Number of consecutive groups of equal values.
Groups of ones **175** Number of separated blocks made of one or more 1 values.	**Groups of zeros** **175** Number of separated blocks made of one or more 0 values.

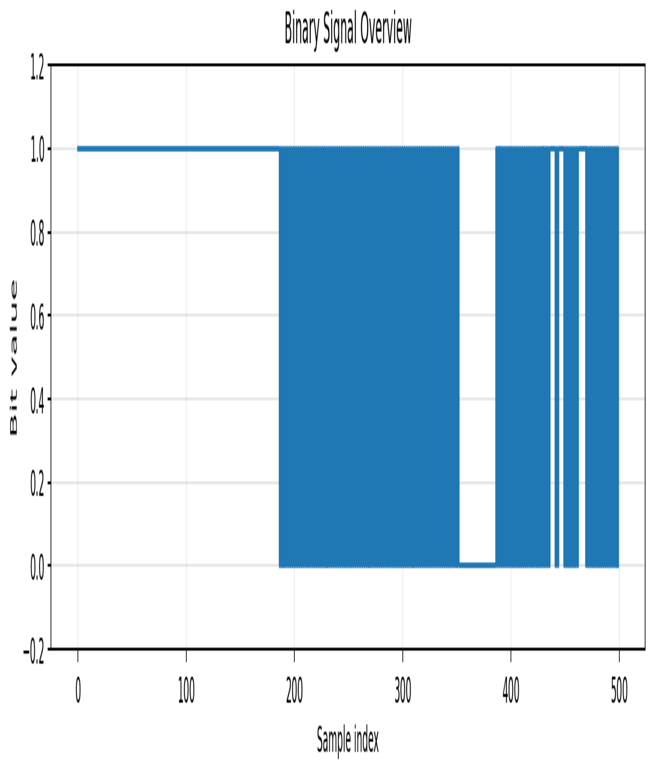

Figure 1. First samples of the binary sequence. The plot is capped for readability when the input is long.

2. Run-Length Structure

Run-length analysis counts consecutive groups of identical values. This is often more readable than a raw distribution plot, especially when long runs are rare or unevenly distributed.

Run Summary

Value	Groups	Mean length	Median length	Min length	Max length
0	175	1.19	1.00	1	34
1	175	2.22	1.00	1	186

Run-Length Matrix

Run length	Count of 0-runs	Count of 1-runs
1	174	168
2	0	0
3	0	0
4	0	0
5	0	4
6	0	0
7	0	2
8	0	0
9	0	0
10	0	0
11	0	0
12	0	0
12+	1	1

Longest Runs

Rank	Value	Start index	End index	Length
1	1	0	185	186
2	0	352	385	34
3	1	462	468	7
4	1	546	552	7
5	1	436	440	5
6	1	444	448	5
7	1	520	524	5
8	1	528	532	5
9	0	186	186	1
10	1	187	187	1

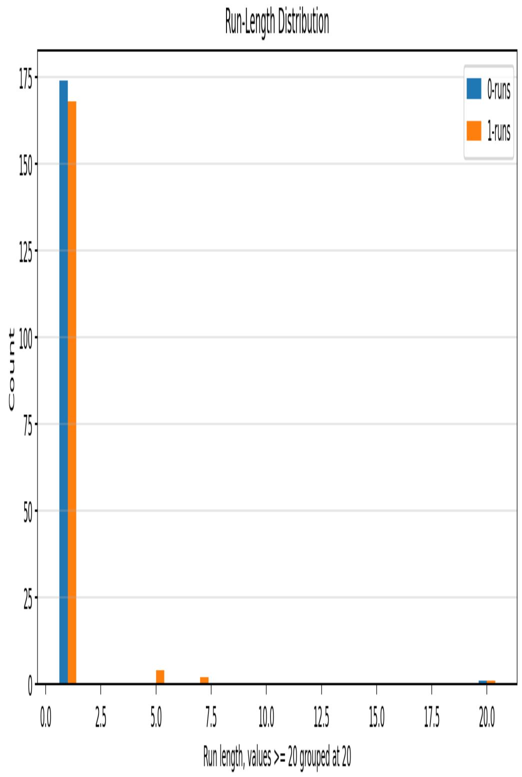

Figure 2. Run-length distribution for zeros and ones. Very long runs are grouped into the last bucket for readability.

3. Transitions & Repeated Blocks

The transition matrix shows how often the sequence moves from one value to another. Repeated block analysis highlights short binary patterns that appear frequently.

Transition Matrix

From / To	0	1	Row total
0	33 (5.55%)	174 (29.24%)	207
1	175 (29.41%)	213 (35.80%)	388
Column total	208	387	595

Most Frequent 4-Bit Blocks

Rank	Block size 4	Occurrences	Frequency
1	1111	199	33.56%
2	1010	167	28.16%
3	0101	166	27.99%
4	0000	31	5.23%
5	1110	7	1.18%
6	1101	7	1.18%

7	1011	6	1.01%
8	0111	6	1.01%
9	0100	1	0.17%
10	1000	1	0.17%
11	0001	1	0.17%
12	0010	1	0.17%

Most Frequent 8-Bit Blocks

Rank	Block size 8	Occurrences	Frequency
1	11111111	179	30.39%
2	10101010	153	25.98%
3	01010101	152	25.81%
4	00000000	27	4.58%
5	11111010	7	1.19%
6	11110101	7	1.19%
7	10101111	6	1.02%
8	01011111	6	1.02%
9	11101010	5	0.85%
10	11010101	5	0.85%
11	10101011	4	0.68%
12	01010111	4	0.68%

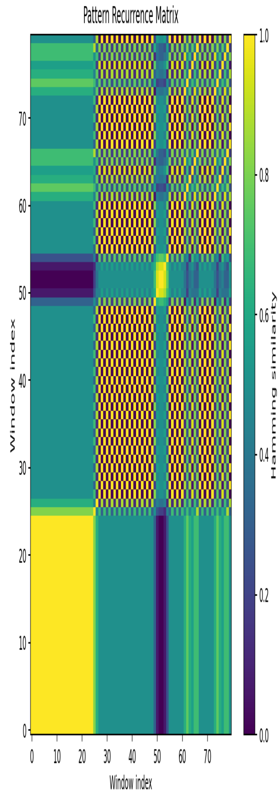

Figure 3. Pattern recurrence matrix. Brighter areas indicate higher similarity between local binary windows.

4. Frequency & Periodicity Analysis

FFT and autocorrelation are complementary. FFT highlights dominant frequencies, while autocorrelation highlights repeated structures in sample-lag space.

Metric	Value	Interpretation
Dominant FFT frequency	0.498322	Cycles per sample. Higher values indicate faster alternation.
Estimated FFT period	2.01	Approximate samples per repeated cycle.
Dominant autocorrelation lag	2	Strongest non-zero lag detected from autocorrelation peaks.
Dominant autocorrelation value	0.9417	Normalized correlation at the dominant lag.

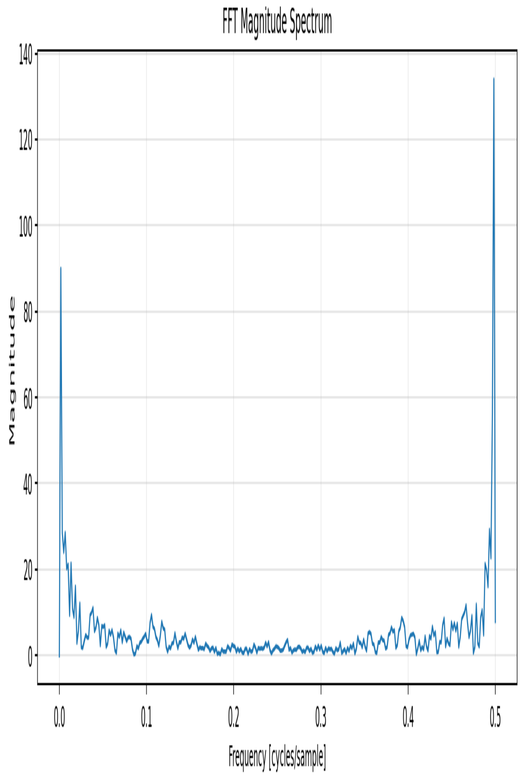

Figure 4. FFT magnitude spectrum after removing the mean/DC component.

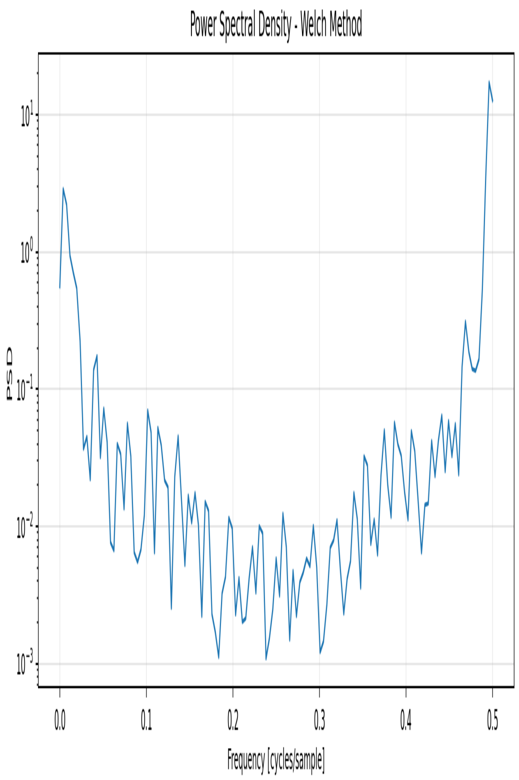

Figure 5. Welch power spectral density. Segment size used: 256 samples.

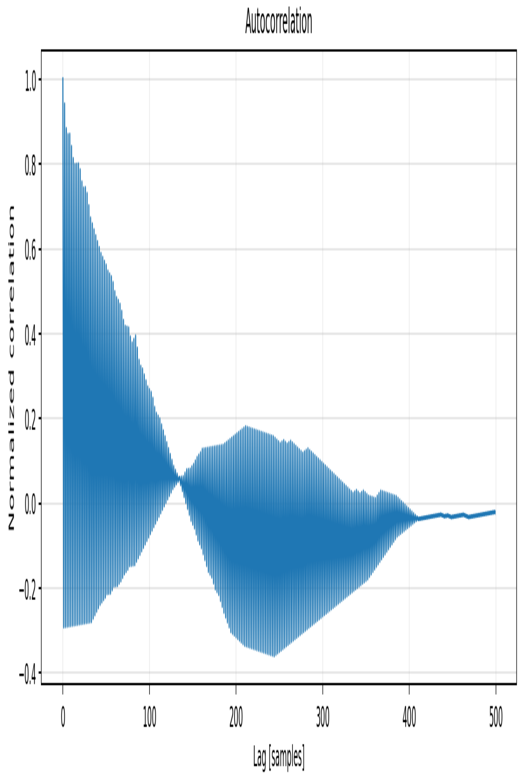

Figure 6. Normalized autocorrelation. Peaks at non-zero lags suggest repeated structure.

5. Wavelet Analysis

Wavelet analysis helps detect structures that vary over the sequence. Unlike FFT, which summarizes global frequency content, wavelets preserve approximate location information.

Metric	Value	Interpretation
Wavelet	Morlet	Good general-purpose continuous wavelet for oscillatory structures.
Strongest scale	62	Scale with the highest average wavelet magnitude.
Scale energy	0.4611	Average magnitude at the strongest scale.

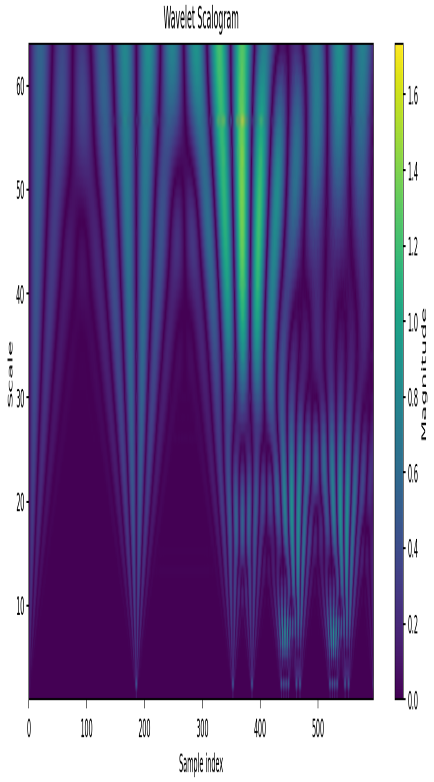

Figure 7. Wavelet scalogram. Strong horizontal bands may indicate persistent scale-level structures.

6. Interpretation & Conclusions

• The sequence shows a strong imbalance between zeros and ones.

• Entropy is moderate, suggesting some structure or symbol bias may exist.

• The sequence contains 175 groups of ones and 175 groups of zeros. The longest run of ones has length 186, while the longest run of zeros has length 34.

• FFT suggests a dominant period of approximately 2.01 samples. Autocorrelation highlights lag 2 as the strongest non-zero repeated structure.

• The recurrence matrix shows low average similarity between windows, suggesting less obvious local repetition.

Recommended next step: compare this report against a known baseline sequence. The most valuable conclusions usually come from comparing entropy, run-lengths, dominant lags and recurrence structure across multiple captures or generated samples.